

Part A

A₁ Write your chosen project title. Then write one main user action that must travel from phone/browser to cloud. Example submit complaint, upload image, get route, predict disease risk.

project title: food trace cloud supply chain Tracker with QR provenance and Spoilage prediction.

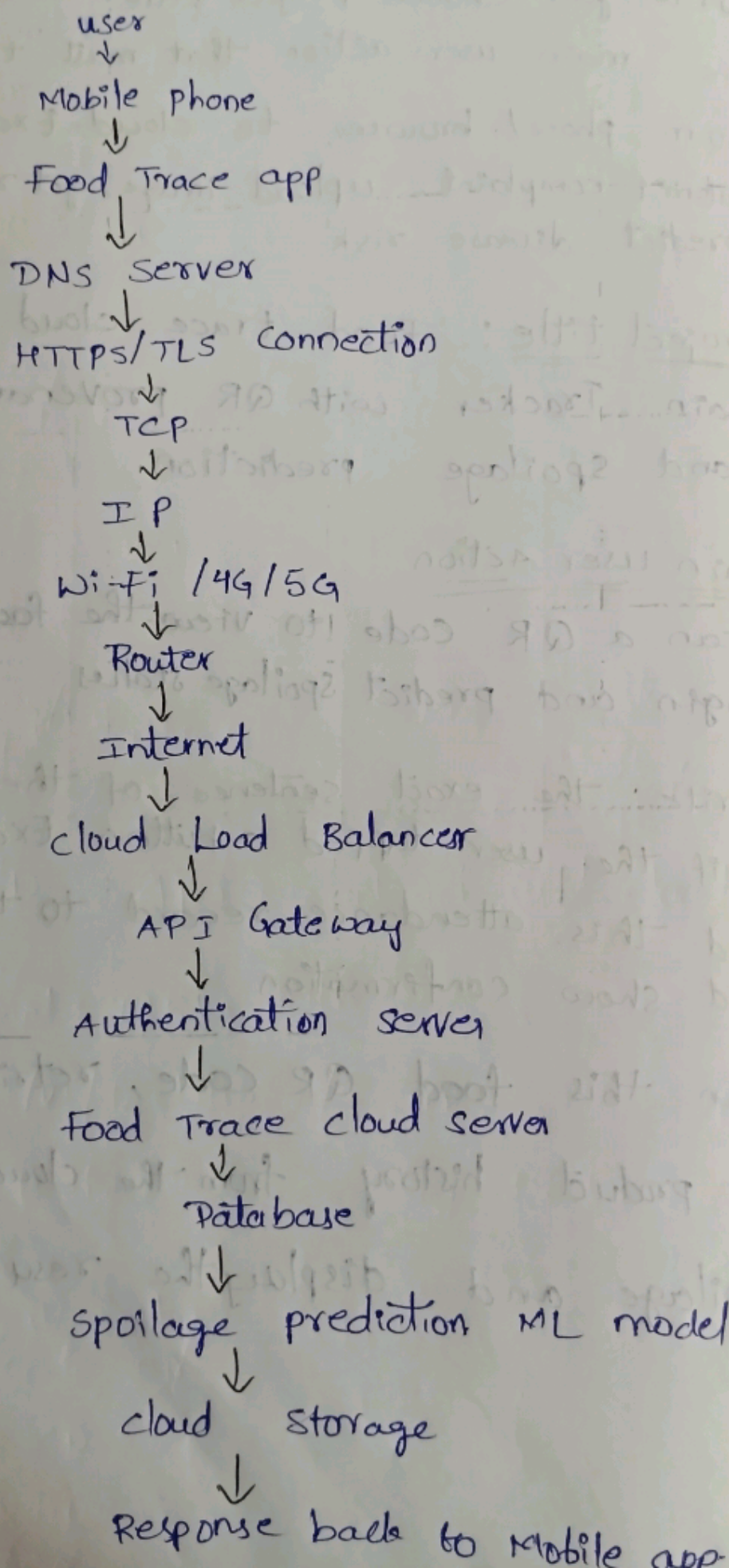
Main User Action

Scan a QR code to view the food products origin and predict spoilage status

A₂ Write the exact sentence of the request as if the user tapped a button. Example
send this attendance record to the cloud and show confirmation
Scan this food QR code, retrieve the product history from the cloud. predict Spoilage and display the result.

Part B - Network journey: phone to cloud

B1 Message path (Draw with arrows)



Layer / component	What it does in your app request	Possible failure	cloud-side protection
DNS	converts food Trace website name into an IP address	DNS server unavailable	Backup DNS Servers
HTTPS / TLS	Encrypts QR Scan data sent to the cloud	invalid SSL Certificate	SSL / TLS encryption
TCP / IP	Delivers data Packets reliably to the cloud	Packet loss or timeout	Packet retransmission
Router / ISP / internet	Transfers request from mobile to cloud	internet / network failure	Redundant network paths
API Gateway / Load Balancer	Receives request and forward it to cloud servers	High traffic / server overload	Load balancing & auto scaling

Part C - cloud Landing zone

cloud Need	Your component choice	Service model	Justification
user identity / Login	Fire base Authentication	SaaS	Secure user login and authentication
App backend / business logic	cloud functions / App server	paas	Process QR Scan and Spoilage prediction
Database / records	Cloud SQL / Firestore	Paas	Stores product supplier and tracking data.
File/image Storage	cloud storage	paas	Stores QR images and food records
Monitoring / analytics	cloud monitoring	SaaS	Monitors application performance and errors.

Part D - API contract and cloud behaviour

Endpoint URL Path

/api/v1/food trace / scan

HTTP Method

POST

Request Data

QR code ID

User ID

Scan Time

Location

Product Name

Origin

Supply chain History

Spoilage prediction

Expiry date

Status

Authentication Rule

JWT Token or user login Authentication

Error case

Invalid QR code

Network Error

Authentication failed

Server unavailable

Part E Elasticity, scalability and Measured

Service

E1 calculate peak Requests / Minute

Given

Total requests = 7,200

Time = 10 minutes

Peak requests / Minute = $7,200 \div 10 = 720$ requests / minute

E2 Calculate Required capacity After spare Capacity.

Given

• keep 25% spare capacity

Required capacity = $720 \times 1.25 = 900$ requests / min

$\therefore 900$ requests / minute.

E3 calculate minimum Number of instances

Given
• one cloud instance handles 180 requests/min

instances required = $900 \div 180 = 5$

5 cloud instances

E4 Elastic/scalable Design Decision

use auto scaling to add or remove cloud servers on traffic. use a load balancer to distribute requests and cloud storage/
database replication for reliable performance

E5 Two Measured-service metrics

- 1. Request Count (requests/minute)
- 2. Average Response Time (Latency)

Part F - AI / Google Research Evidence

Search/ prompt used	Tool used	Useful Finding	How I applied it
How does API Gateway work in cloud computing	ChatGPT, Gemini, cloud AI	API Gateway securely routes client requests to find services	Used API Gateway to process QR scan requests securely